FFAF30UA60S

30 A, 600 V, Ultrafast II Single Diode

The FFAF30UA60S is an Ultrafast ll dual diode with low forward voltage drop and rugged UIS capability. This device is intended for use as freewheeling and clamping diodes in a variety of switching power supplies and other power switching applications. It is specifically suited for use in switching power supplies and industrial application as welder and UPS application.

Features

- Ultrafast Recovery, T_{rr} < 90 ns (@ I_F = 30 A)
- Max Forward Voltage, V_F = 2.2 V (@ T_C = 25°C)
- 600 V Reverse Voltage and High Reliability
- Avalanche Energy Rated
- These Devices are Pb−Free and are RoHS Compliant

Typical Applications

- Boost Diode in PFC and SMPS
- Welder, UPS, and Motor Control Application

ABSOLUTE MAXIMUM RATINGS

Per leg at T_C = 25°C unless otherwise noted

Rating	Symbol	Value	Unit
Peak Repetitive Reverse Voltage	V_{RRM}	600	V
Working Peak Reverse Voltage	V_{RWM}	600	V
DC Blocking Voltage	V_R	600	V
Average Rectified Forward Current (Rated V_R, T_C = 45°C)	$I_{F(AV)}$	30	A
Non−repetitive Peak Surge Current 60 Hz Single Half−Sine Wave	I_{FSM}	180	A
Operating and Storage Temperature Range	T_J, T_{STG}	−65 to +175	°C

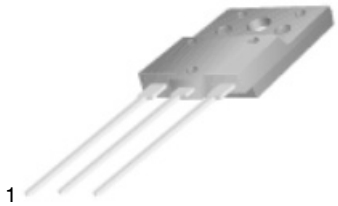
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

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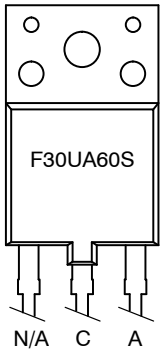
30 A, 600 V ULTRAFAST II RECTIFIER





TO−3PF
CASE 340AH

MARKING DIAGRAM



F30UA60S = Specific Device Code

ORDERING INFORMATION

Device	Package	Shipping
FFAF30UA60S	TO−3PF	30 / Rail

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THERMAL CHARACTERISTICS

Symbol	Parameter	Value	Unit
$R_{\theta JC}$	Thermal Resistance, Junction−to−Case, Steady State (Assumes 600 mm^2 1 oz. copper bond pad, on an FR4 board)	2.4	°C/W

ELECTRICAL CHARACTERISTICS

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
V_{FM}1	Instantaneous Forward Voltage	I_F = 30 A, T_C = 25°C I_F = 30 A, T_C = 125°C	– –	– –	2.2 2.0	V
I_{RM}1	Instantaneous Reverse Current	V_R = 600 V, T_C = 25°C V_R = 600 V, T_C = 125°C	– –	– –	100 150	μA
Trr	Reverse Recovery	I_F = 30 A, di_F/dt = 200/μs, T_C = 25°C	–	–	90	Ns
Irr			–	–	8	A
Qrr			–	–	360	nC
W_{AVL}	Avalanche Energy	L = 40 mH	20	–	–	mJ

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse: Test Pulse width = 300 μs, Duty Cycle = 2%

Test Circuit and Waveforms

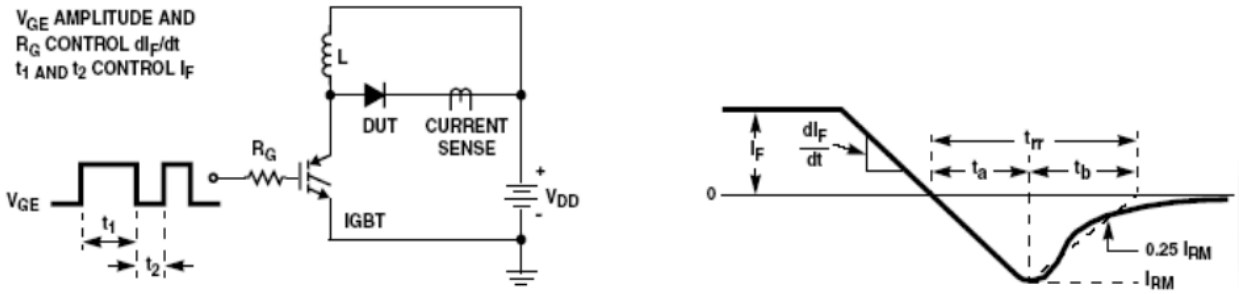


Figure 1. Diode Reverse Recovery Test Circuit & Waveform

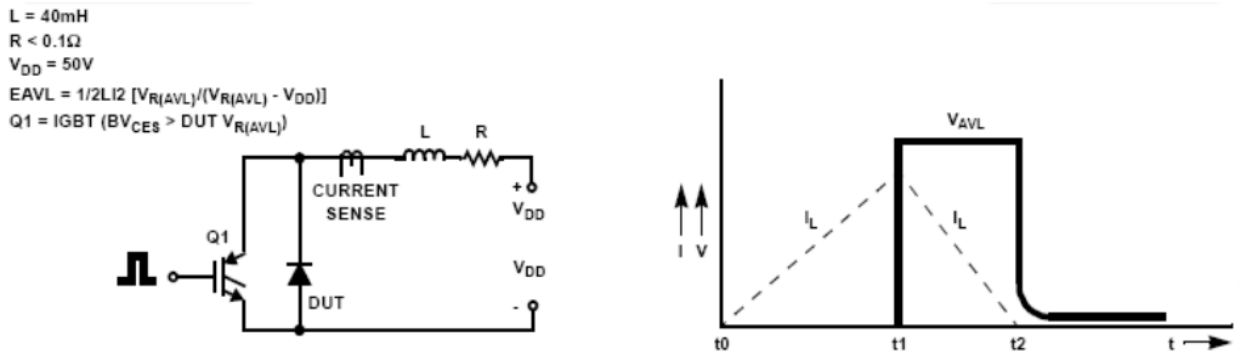


Figure 2. Unclamped Inductive Switching Test Circuit & Waveform

TYPICAL CHARACTERISTICS

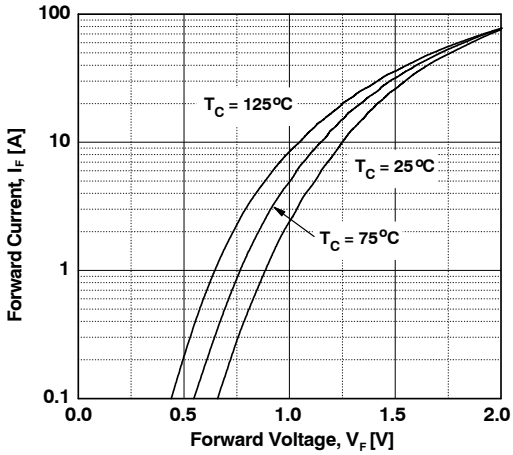


Figure 3. Typical Forward Voltage Drop vs. Forward Current

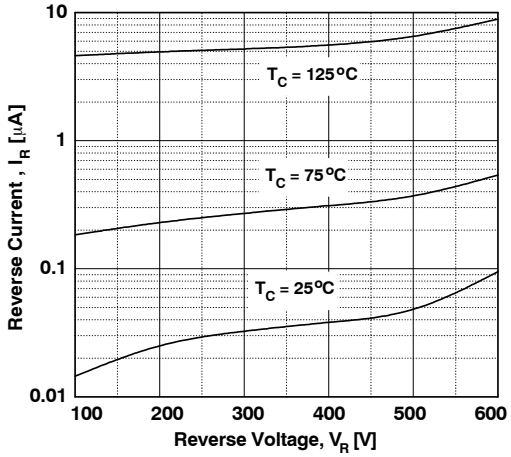


Figure 4. Typical Reverse Current vs. Reverse Voltage

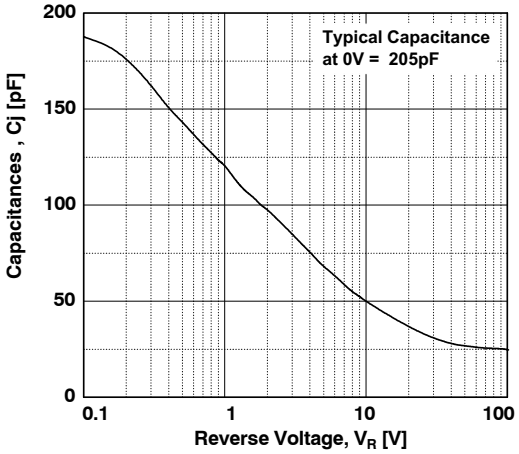


Figure 5. Typical Junction Capacitance

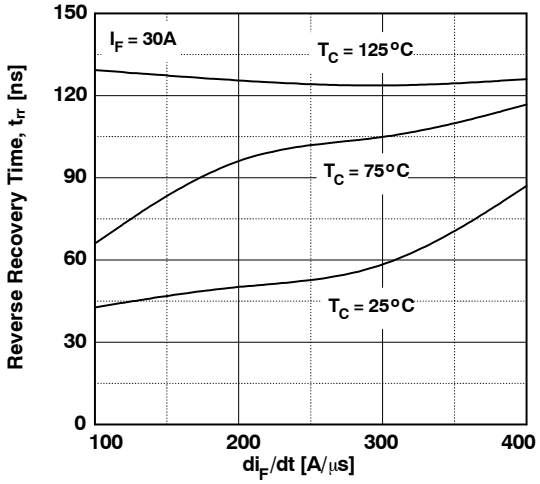


Figure 6. Typical Reverse Recovery Time vs. di_F/dt

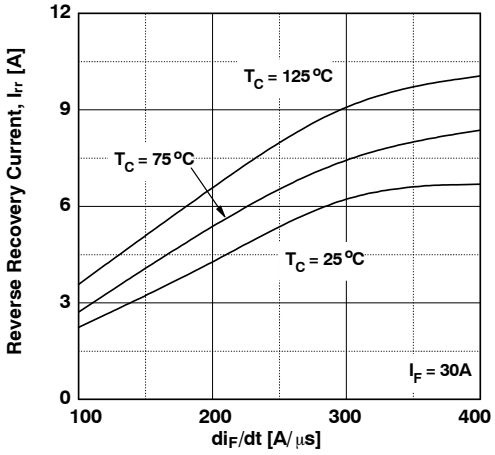


Figure 7. Typical Reverse Recovery Current vs. di_F/dt

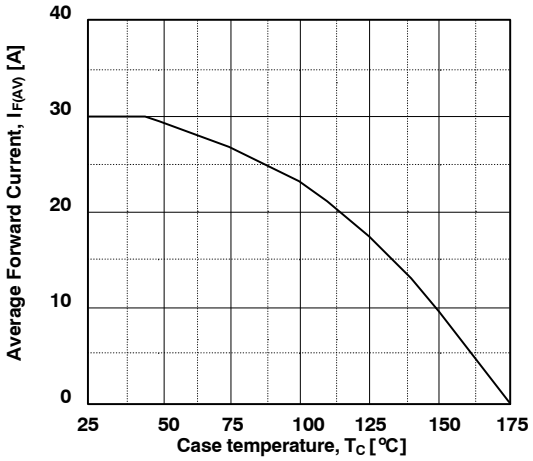
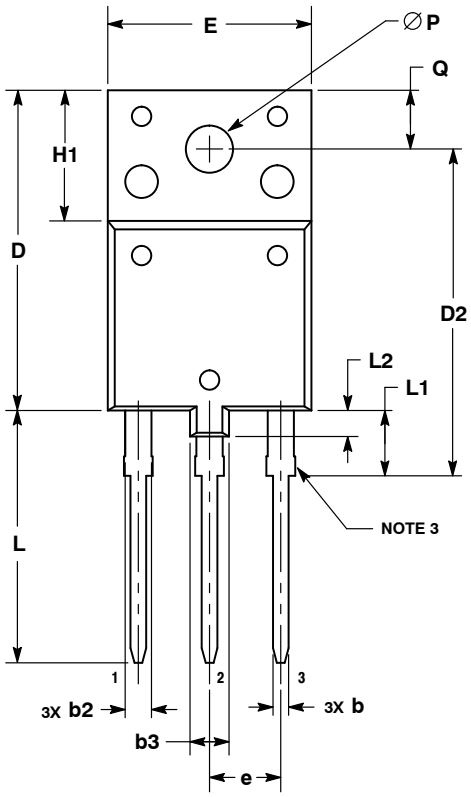
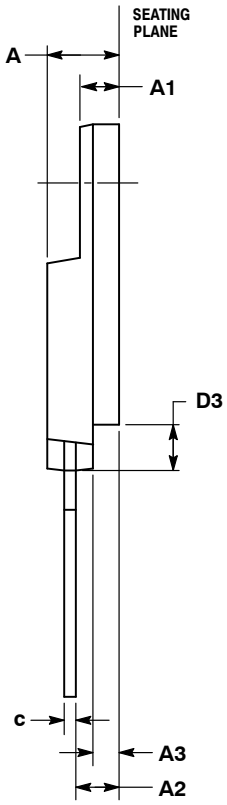


Figure 8. Forward Current Derating Curve

TO−3PF−3L
CASE 340AH
ISSUE A

DATE 09 JAN 2015





NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2009.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. CONTOUR UNCONTROLLED IN THIS AREA (6 PLACES).
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH OR GATE PROTRUSIONS. MOLD FLASH AND GATE PROTRUSIONS NOT TO EXCEED 0.13 PER SIDE. THESE DIMENSIONS ARE TO BE MEASURED AT THE OUTERMOST EXTREME OF THE PLASTIC BODY.
5. DIMENSION b2 DOES NOT INCLUDE DAMBAR PROTRUSION. LEAD WIDTH INCLUDING PROTRUSION SHALL NOT EXCEED 2.20.

	MILLIMETERS	
DIM	MIN	MAX
A	5.30	5.70
A1	2.80	3.20
A2	3.10	3.50
A3	1.80	2.20
b	0.65	0.95
b2	1.90	2.15
b3	3.80	4.20
c	0.80	1.10
D	24.30	24.70
D2	24.70	25.30
D3	3.30	3.70
E	15.30	15.70
e	5.35	5.55
H1	9.80	10.20
L	19.10	19.50
L1	4.80	5.20
L2	1.90	2.20
P	3.40	3.80
Q	4.30	4.70

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